นางสาวณัฐนันทน์ บุษดี 673380037-1 Section 1

TCP and UDP Communications

File :
https://drive.google.com/file/d/1XKONELVi64eprPiMl2WCpt0LMkEa05Yp/view?usp=sharing

Objectives

Part 1: Generate Network Traffic in Simulation Mode

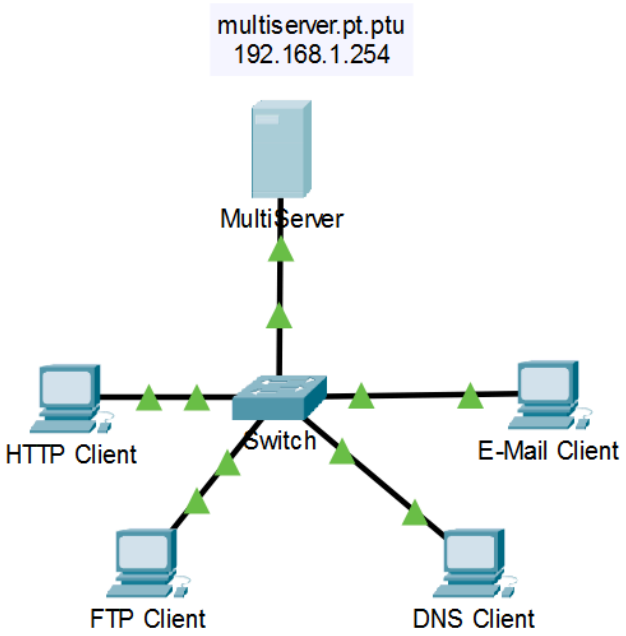
Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding TCP and UDP in detail. Packet Tracer simulation mode provides you the ability to view the state of different PDUs as they travel through the network.

Packet Tracer Simulation mode enables you to view each of the protocols and the associated PDUs. The steps outlined below lead you through the process of requesting network services using various applications that are available on a client PC. You will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested the data or is sending the data. Packet Tracer will not score this activity.

Instructions



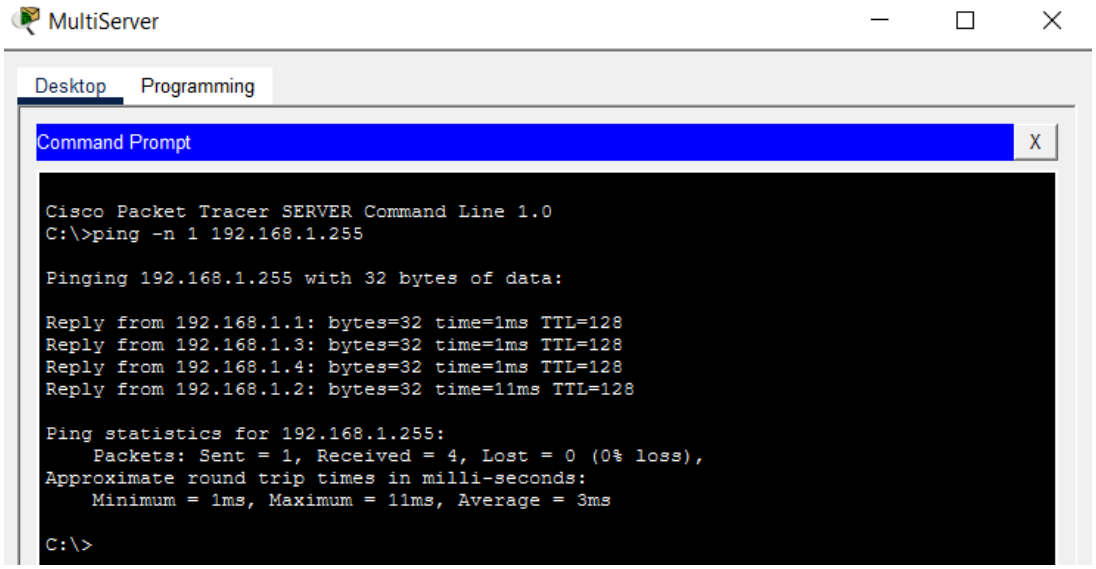
ทุกเครื่องต่อสายเข้ากับ Switch และมี DNS Server = 192.168.1.254

Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

ไปที่ Simulation

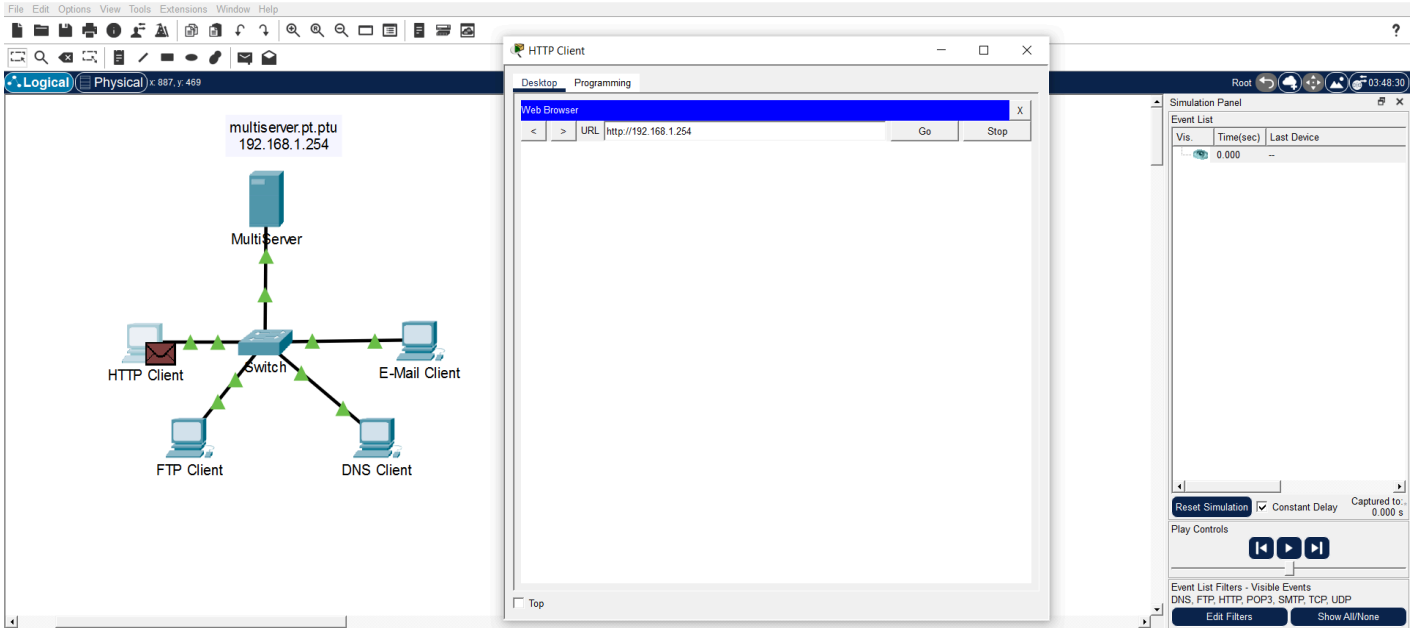
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

คลิก **MultiServer** และไปที่ **Desktop → Command Prompt**
พิมพ์ ping -n 1 192.168.1.255



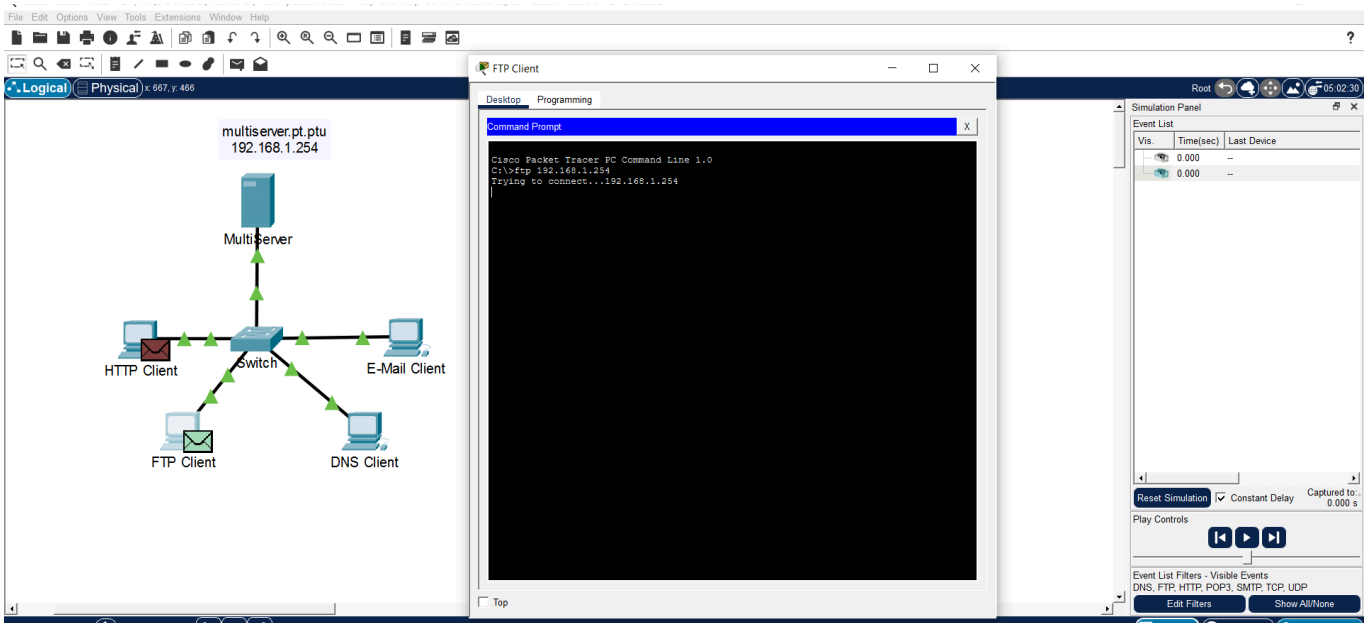
Step 2: Generate web (HTTP) traffic.

คลิก **HTTP Client** และไปที่ Desktop → **Web Browser**
ช่อง URL ใส่ 192.168.1.254 แล้วกด go



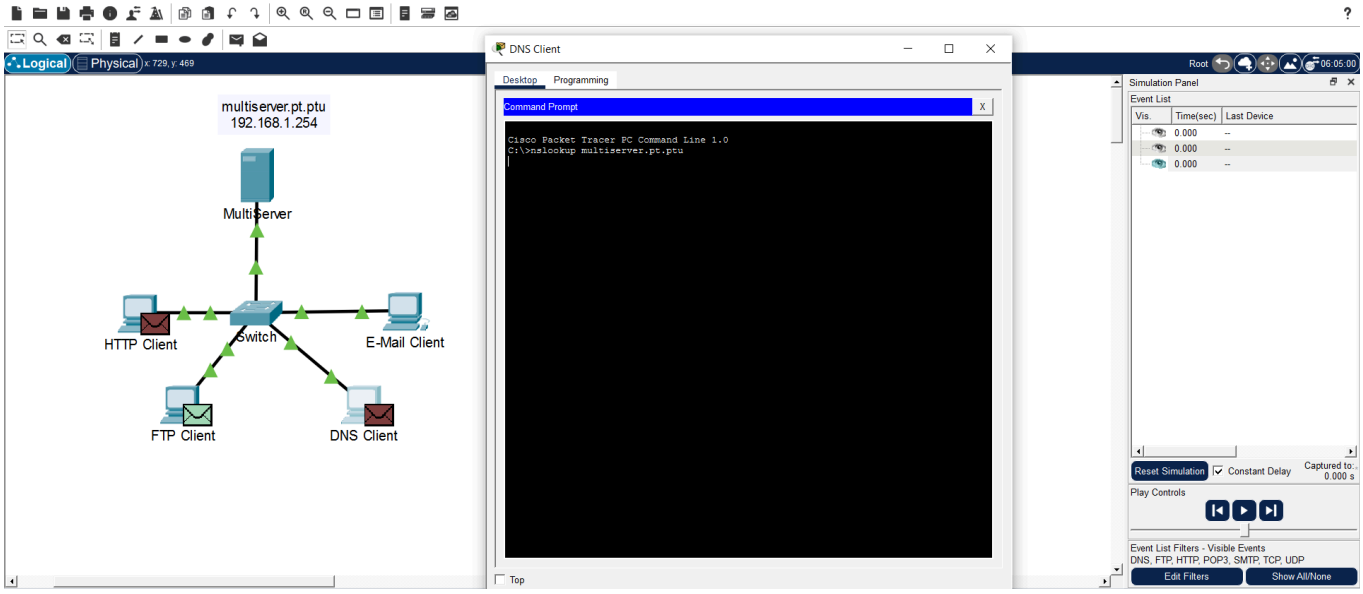
Step 3: Generate FTP traffic.

คลิก **FTP Client** และไปที่Desktop → **Command Prompt**
พิมพ์ ftp 192.168.1.254



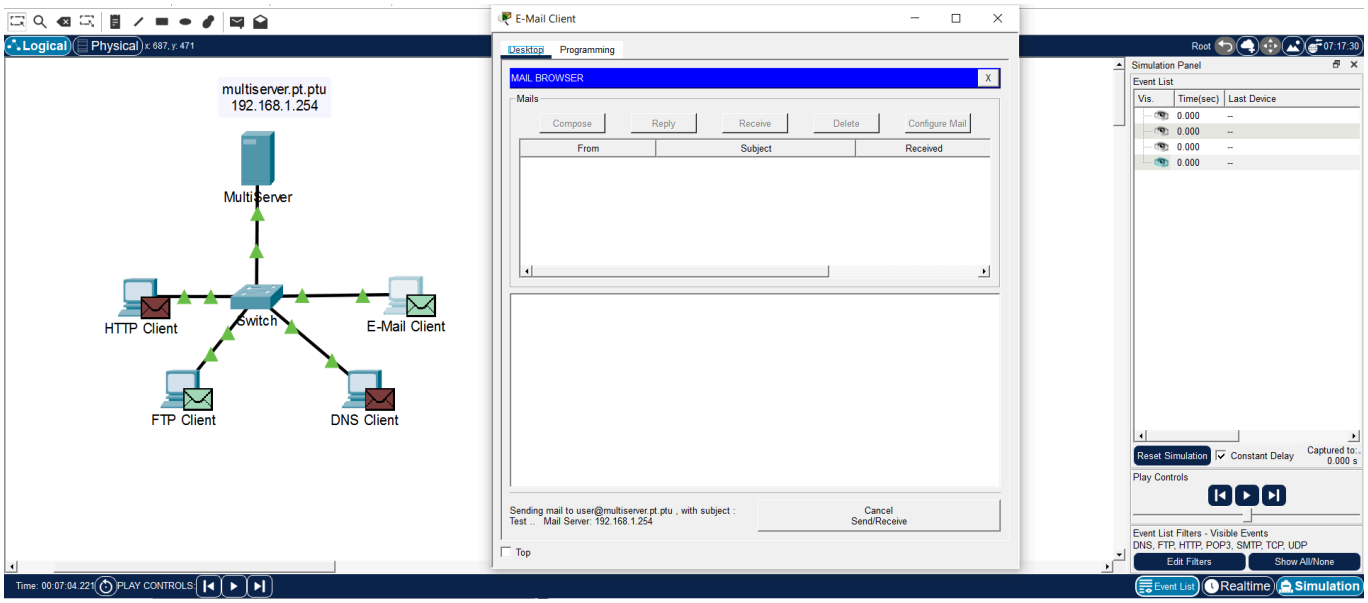
Step 4: Generate DNS traffic.

คลิก **DNS Client** และไปที่Desktop → **Command Prompt**
พิมพ์ nslookup multiserver.pt.ptu

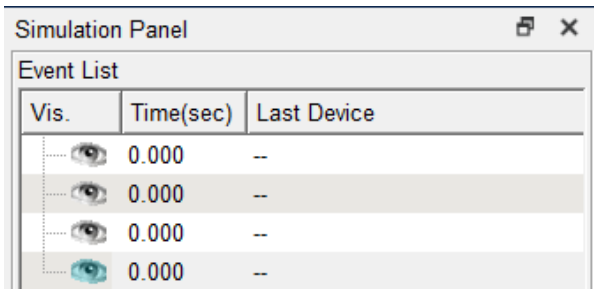


Step 5: Generate Email traffic.

คลิก E-Mail Client ไปที่ Desktop → Email แล้วกด Compose
กรอก
To: user@multiserver.pt.ptu
Subject: -
Body: -
กด Send

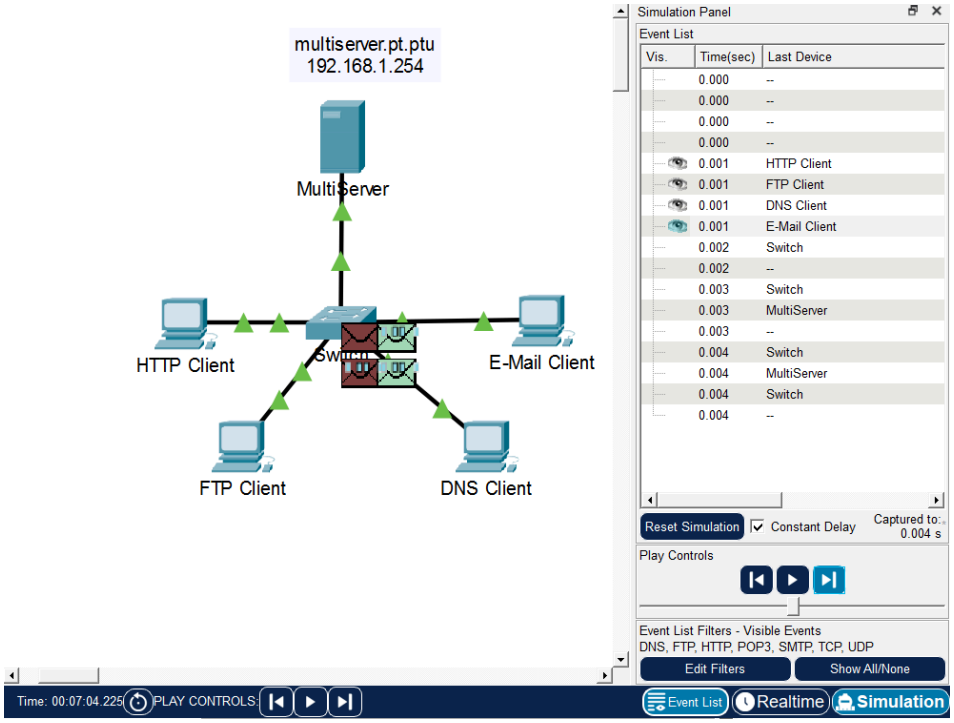


Step 6: Verify that the traffic is generated and ready for simulation.

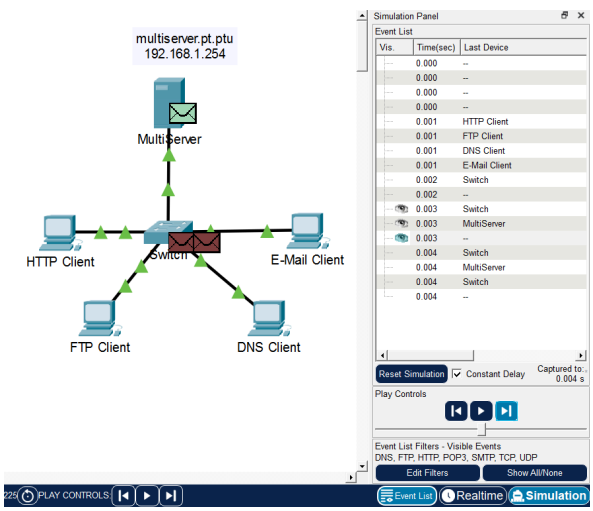


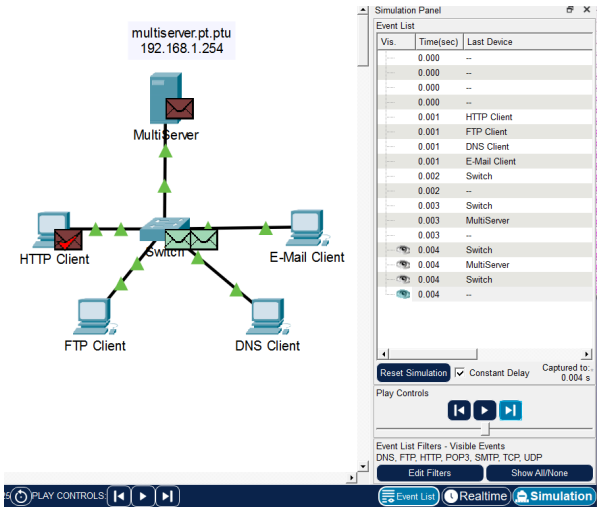
Step 7: Examine multiplexing as the traffic crosses the network.

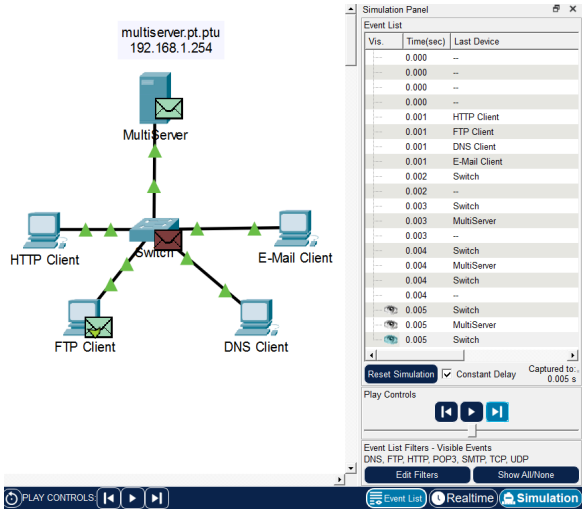
a. Click **Capture/Forward** once. All of the PDUs travel to the switch.

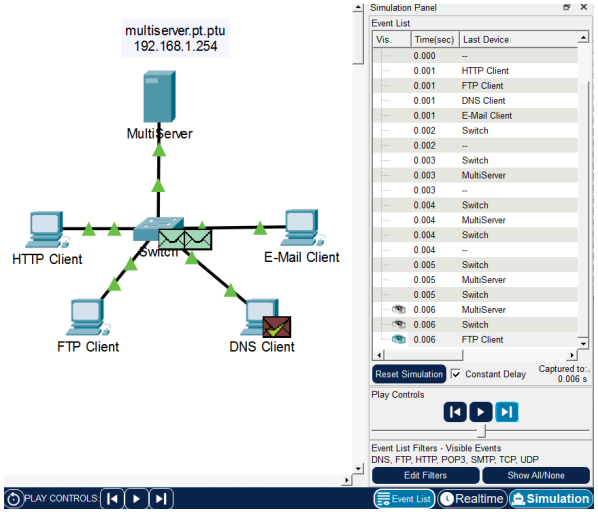


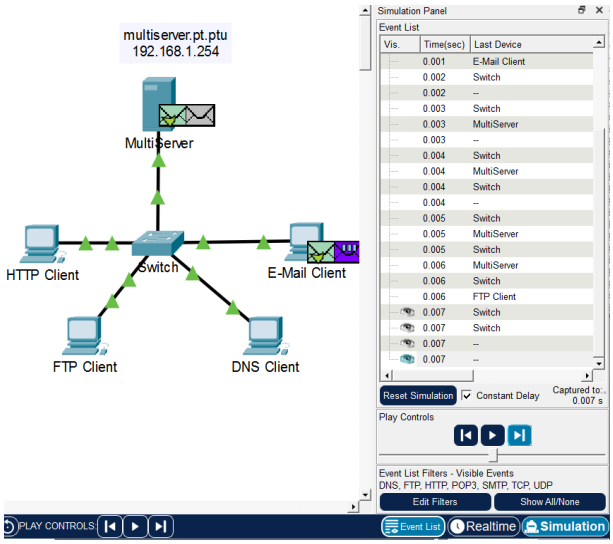
b. Click **Capture/Forward** six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

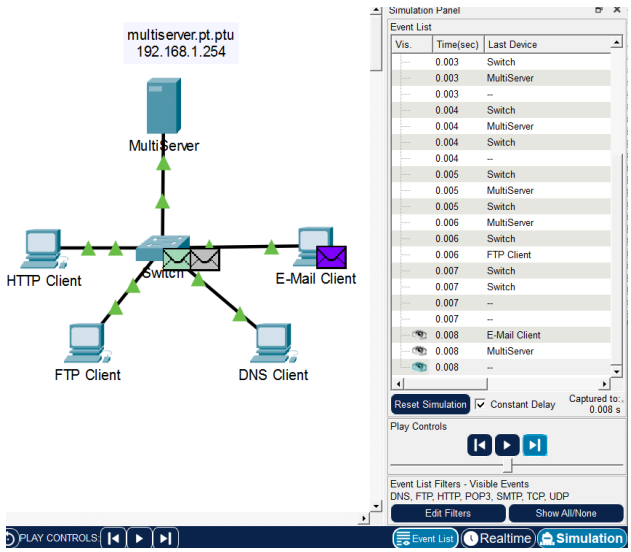












What is this called?

การที่ข้อมูลจากหลายอุปกรณ์สามารถส่งผ่านสายสื่อสารเส้นเดียวกันได้ แต่ส่งทีละชุด เรียกว่า **Multiplexing**

A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

สีที่แตกต่างกันของ PDU ใน Simulation Panel ใช้แทน **ชนิดของโปรโตคอลที่แตกต่างกัน** เช่น

- สีหนึ่งแทน TCP
- สีหนึ่งแทน UDP
- สีหนึ่งแทน HTTP, FTP, DNS หรือ Email

เพื่อช่วยให้ผู้ใช้สามารถแยกแยะและติดตามการทำงานของโปรโตคอลแต่ละชนิดได้ง่ายขึ้นระหว่างการจำลองการทำงานของเครือข่าย

Part 2: Examine Functionality of the TCP and UDP Protocols

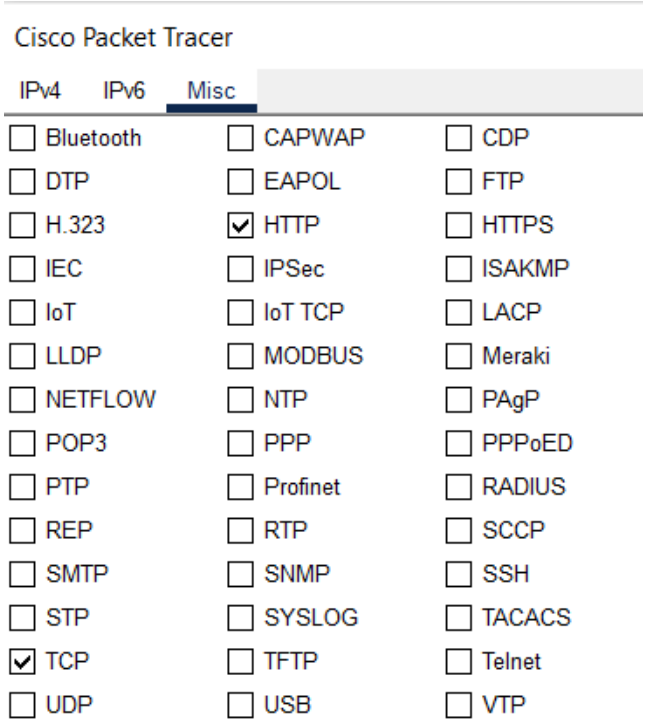
Step 1: Examine HTTP traffic as the clients communicate with the server.

a. Click **Reset Simulation**.

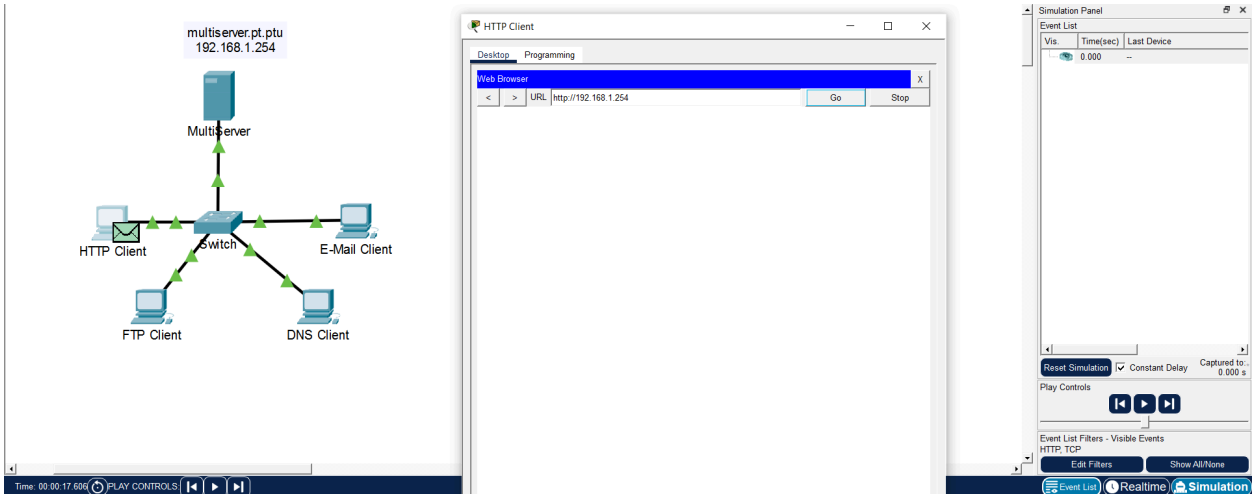
b. Filter the traffic that is currently displayed to only **HTTP** and **TCP** PDUs. To filter the traffic that is currently displayed:

1) Click **Edit Filters** and toggle the **Show All/None** button.

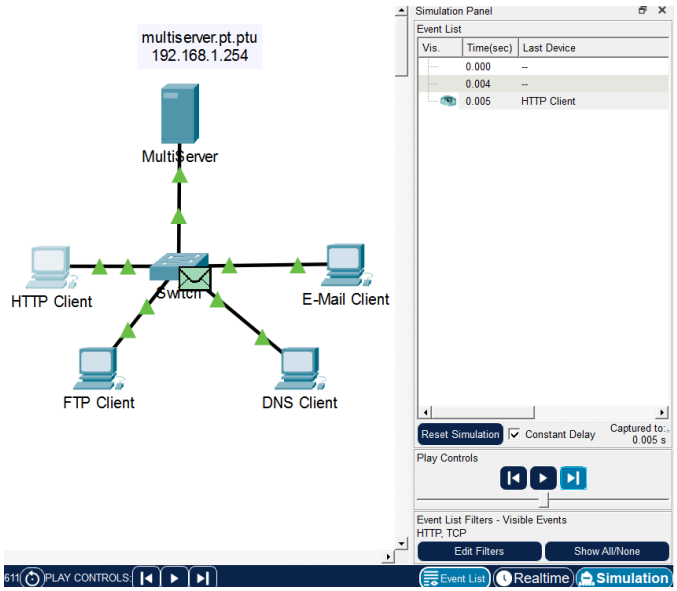
2) Select **HTTP** and **TCP**. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only **HTTP** and **TCP** PDUs.



c. Open the browser on HTTP Client and enter **192.168.1.254** in the URL field. Click **Go** to connect to the server over HTTP. Minimize the HTTP Client window.



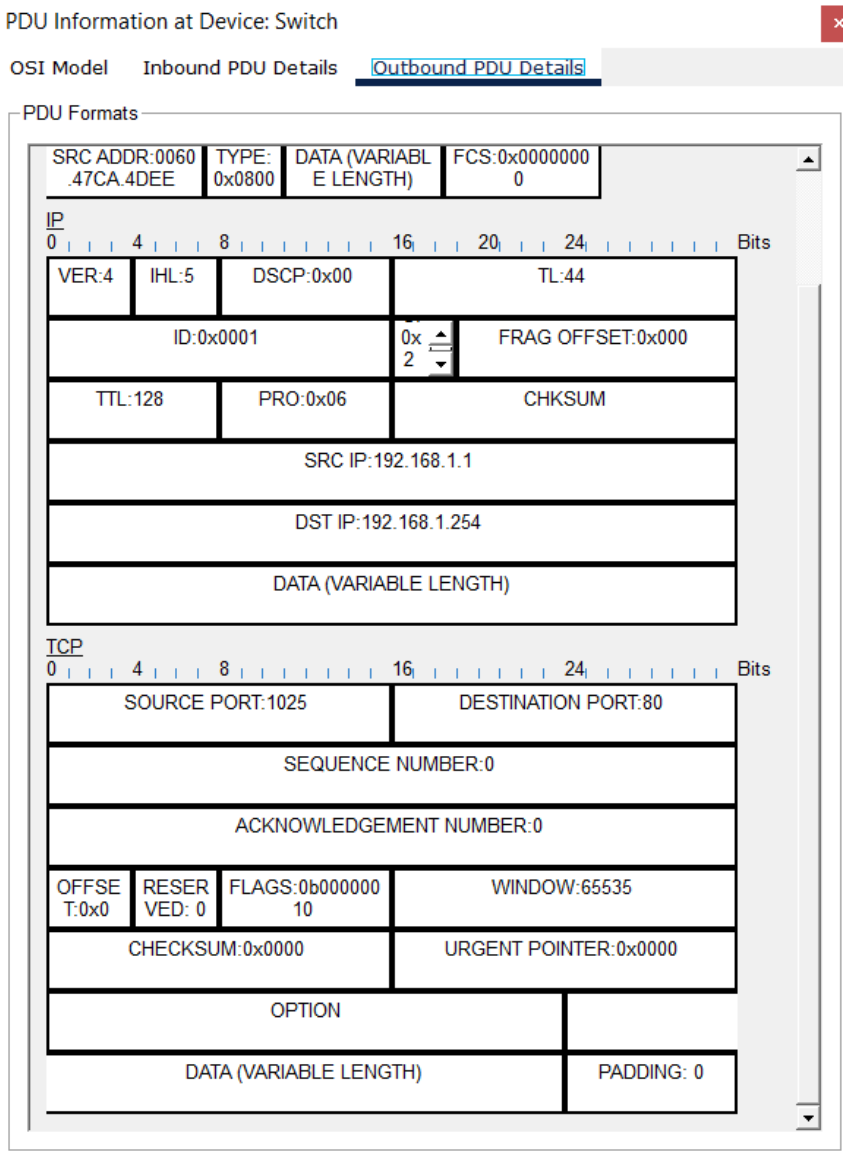
d. Click **Capture/Forward** until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.



Why did it take so long for the HTTP PDU to appear?

เพราะ TCP ต้องมีการ **3-way handshake (SYN, SYN-ACK, ACK)** ก่อนเริ่มส่งข้อมูล HTTP

e. Click the PDU envelope to show the PDU details. Click the **Outbound PDU Details** tab and scroll down to the second to the last section.



Questions:

What is the section labeled?

Transmission Control Protocol (TCP)

Are these communications considered to be reliable?

TCP เป็น reliable protocol

Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

SRC PORT = 1025

DEST PORT = 80

SEQUENCE NUM = 0

ACK NUM = 0

f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

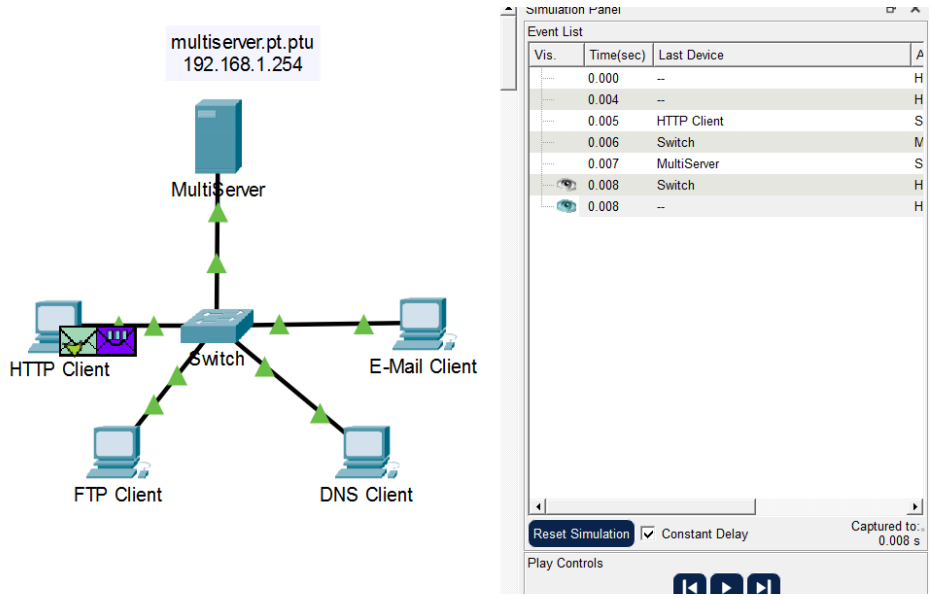
Flag Place	**6**	**5**	**4**	**3**	**2**	**1**
Value	URG	ACK	PSH	RST	SYN	FIN

Flags = 0b00000010

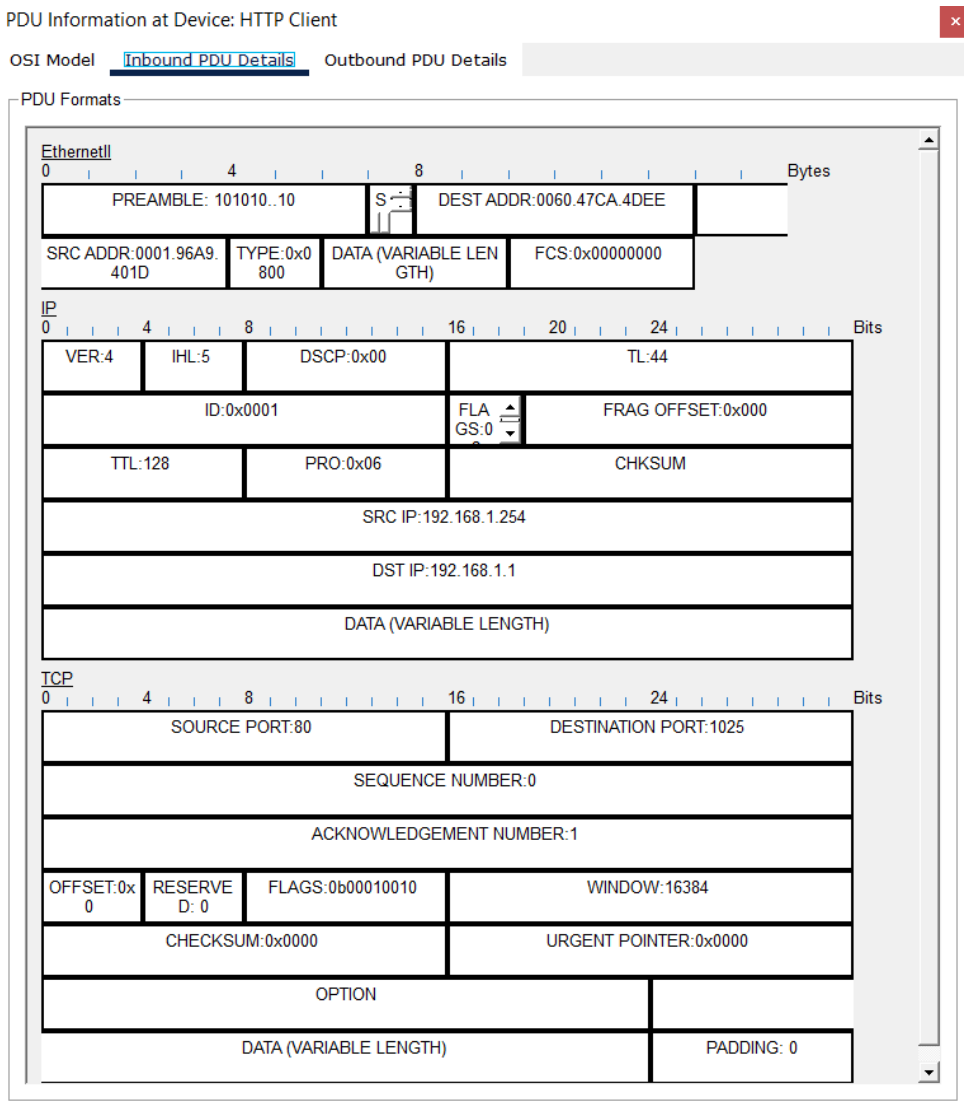
Which TCP flags are set in this PDU?

SYN flag is set (SYN = 1)

g. Close the PDU and click **Capture/Forward** until a PDU with a checkmark returns to the **HTTP Client**.



h. Click the PDU envelope and select **Inbound PDU Details**.

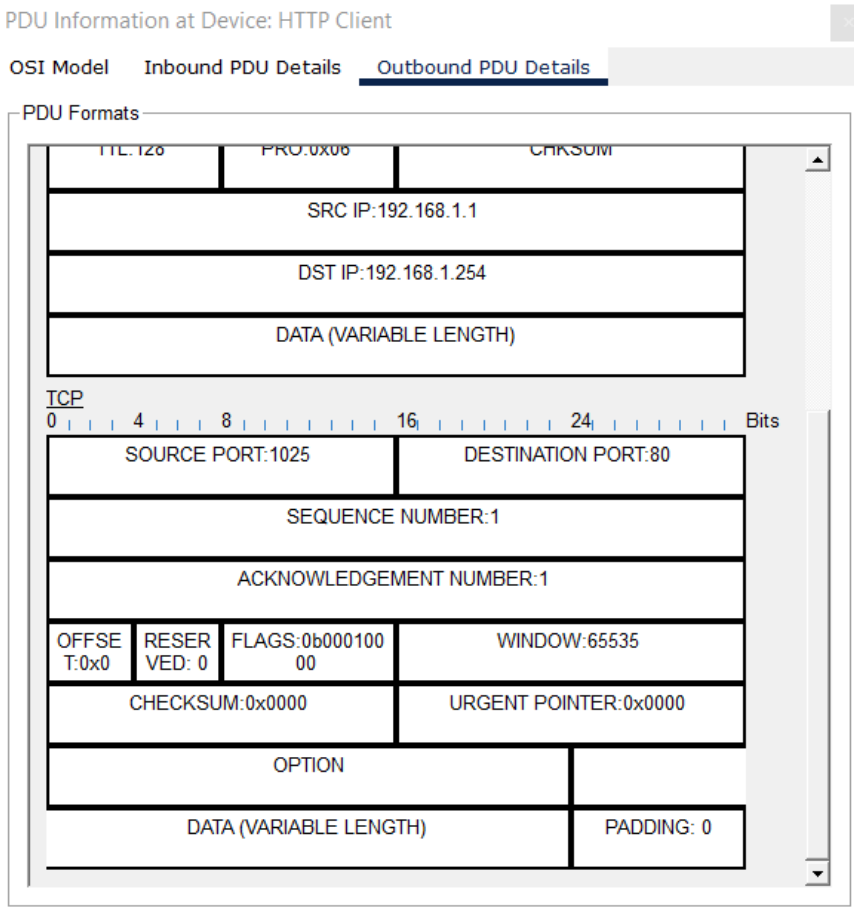


Question:

How are the port and sequence numbers different than before?

<u>ค่าของพอร์ตต้นทางและปลายทางสลับกัน และหมายเลขลำดับ(SEQUENCE NUM)และหมายเลขยืนยันการรับส่งข้อมูล(ACK NUM)ก็เปลี่ยนแปลงไป</u>

i. Click the HTTP PDU which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.



Question:

What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

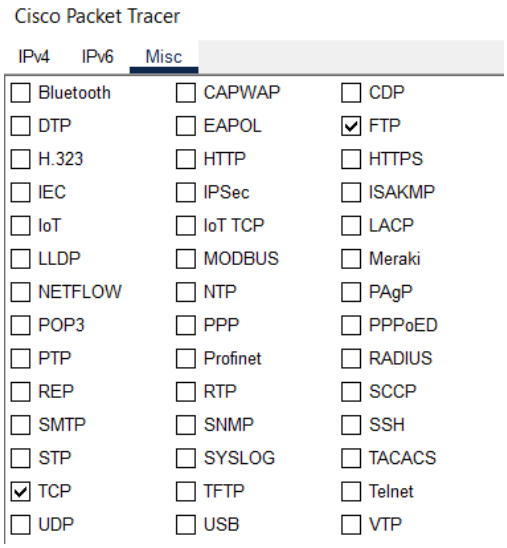
<u>ส่วน TCP ในที่นี้ได้เพิ่มหมายเลขลำดับ(SEQUENCE NUM)และหมายเลขยืนยันสำหรับการถ่ายโอนข้อมูล(ACK NUM) ได้รับการอัปเดต เลข SRC PORT สลับตำแหน่งกับ DEST PORT</u>

j. Reset the simulation.

Step 2: Examine FTP traffic as the clients communicate with the server.

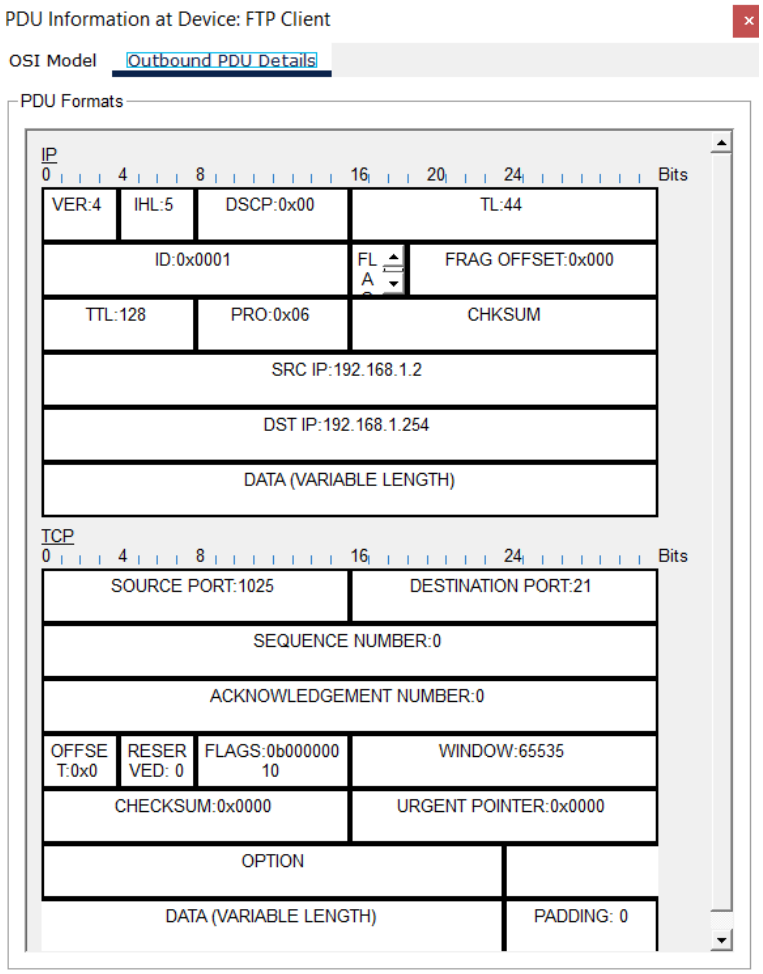
a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering **ftp 192.168.1.254**.

b. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.



c. Click **Capture/Forward**. Click the second PDU envelope to open it.

Click the **Outbound PDU Details** tab and scroll down to the TCP section.



Question:

Are these communications considered to be reliable?

<u>ใช่ FTP ใช้ TCP ซึ่งมีความน่าเชื่อถือ</u>

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

SRC PORT = 1025

DEST PORT = 21

SEQUENCE NUM = 0

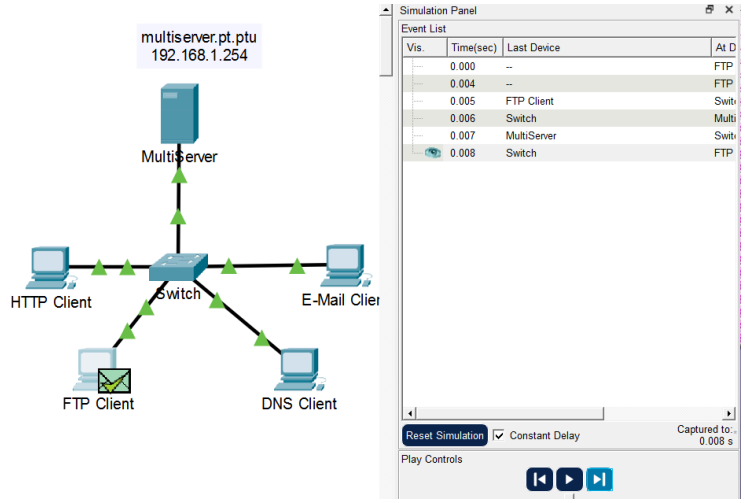
ACK NUM = 0

Question:

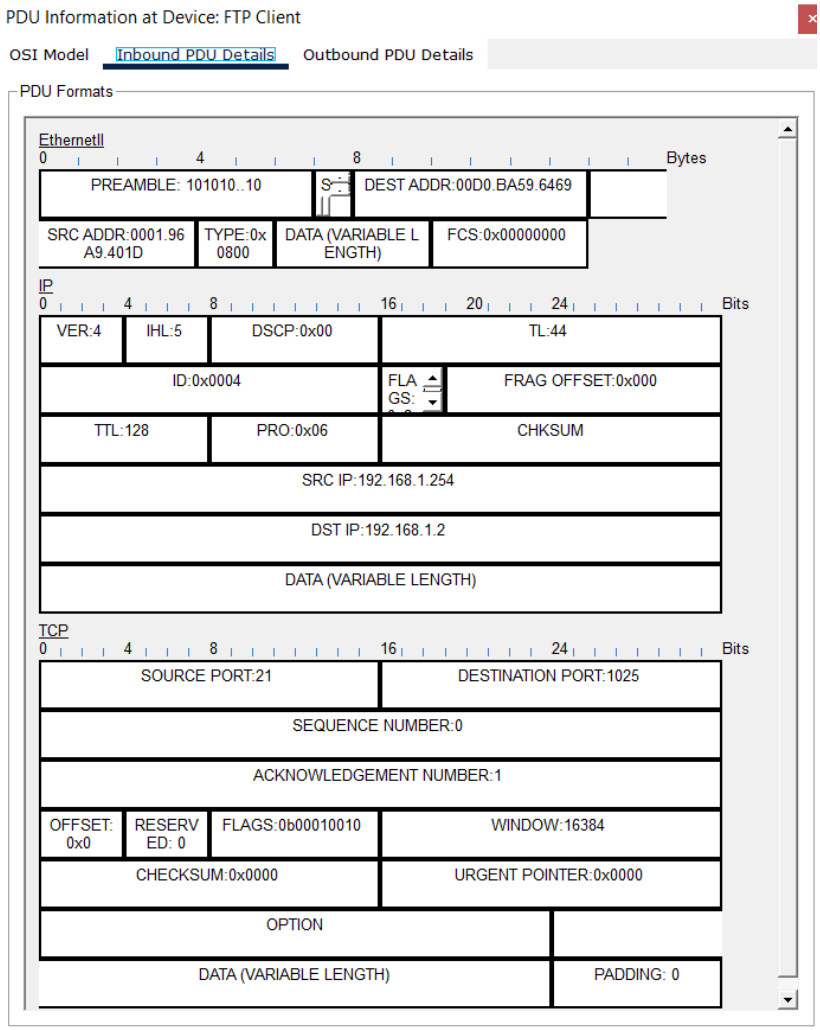
What is the value in the flag field?

<u>มีการตั้งค่าแฟล็ก SYN</u>

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.



f. Click the PDU envelope and select **Inbound PDU Details**.

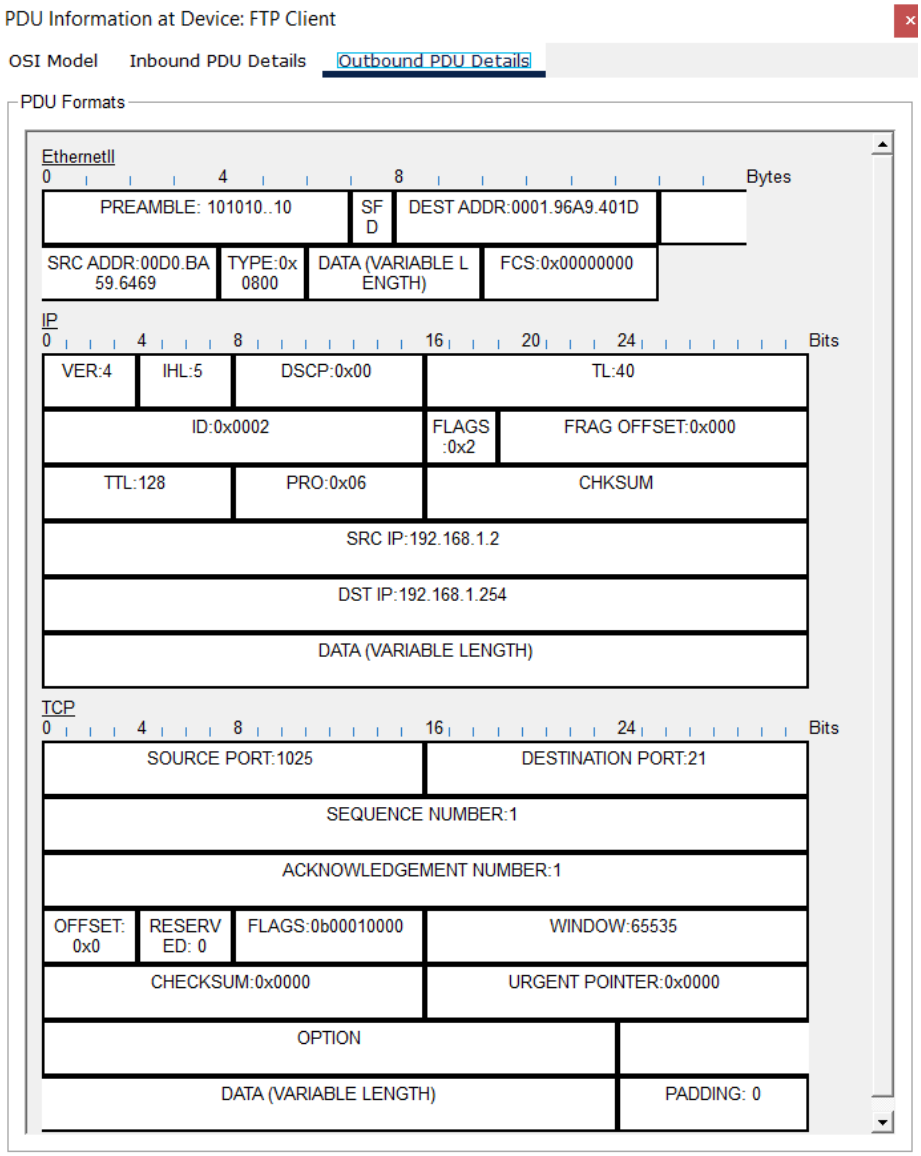


Question:

How are the port and sequence numbers different than before?

<u>ค่าของพอร์ตถูกสลับตำแหน่ง และหมายเลขยืนยันการรับส่งข้อมูล(ACK NUM)ก็เปลี่ยนแปลงไป แต่หมายเลขลำดับ(SEQUENCE NUM) เหมือนเดิม</u>

g. Click the **Outbound PDU Details** tab.



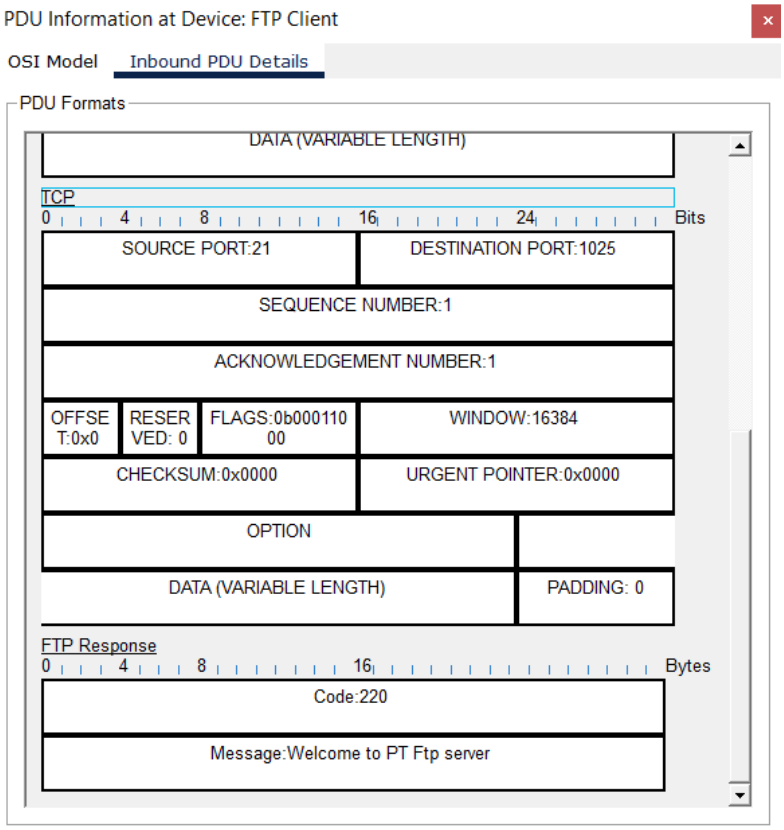
Question:

How are the port and sequence numbers different from the previous results?

<u>ค่าของพอร์ตถูกสลับตำแหน่ง และหมายเลขลำดับ(SEQUENCE NUM)และหมายเลขยืนยันการรับส่งข้อมูล(ACK NUM)ก็เปลี่ยนแปลงไป</u>

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section.



Question:

What is the message from the server?

Welcome to PT Ftp server

j. Click Reset Simulation.

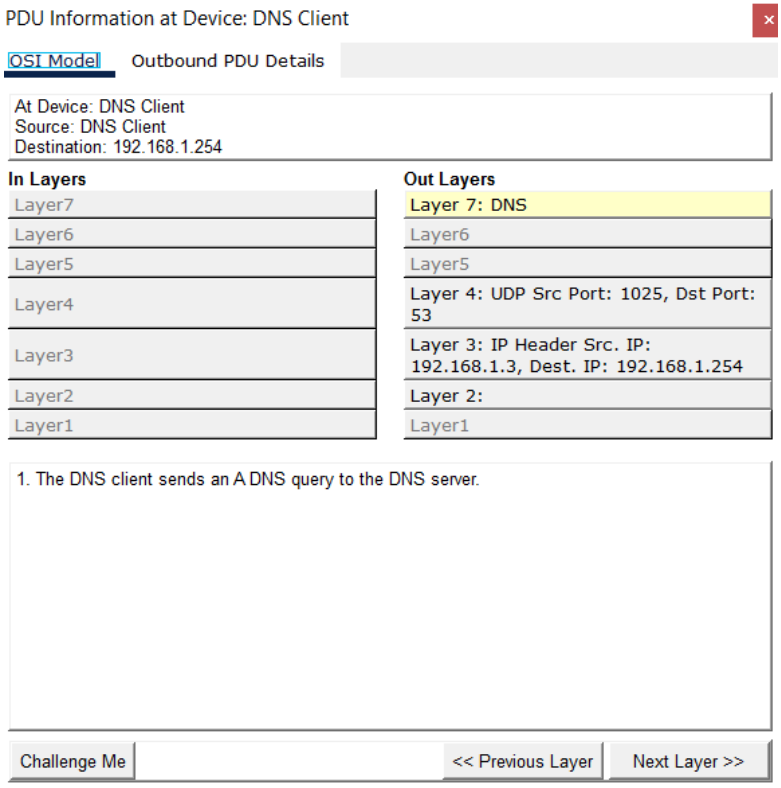
Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.



Question:

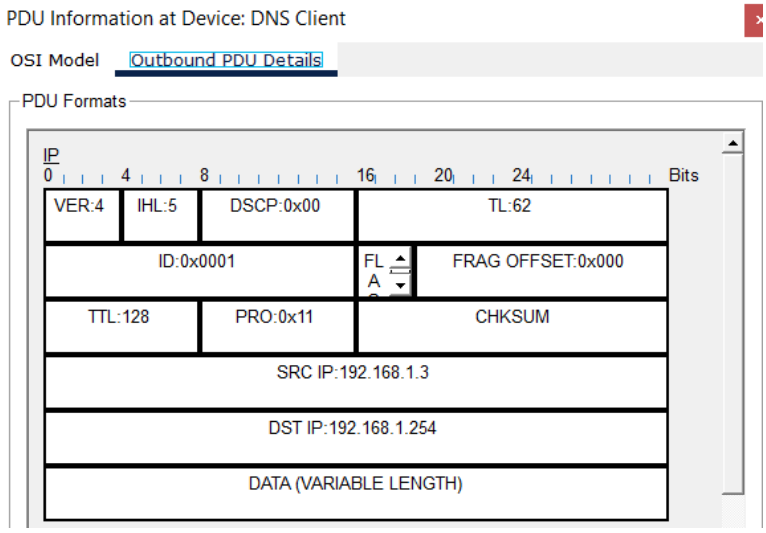
What is the Layer 4 protocol?

UDP Src Port: 1025,Dst Port: 53

Are these communications considered to be reliable?

UDP ไม่น่าเชื่อถือ

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the **SRC PORT** and **DEST PORT** values.



SRC PORT = 192.168.1.3

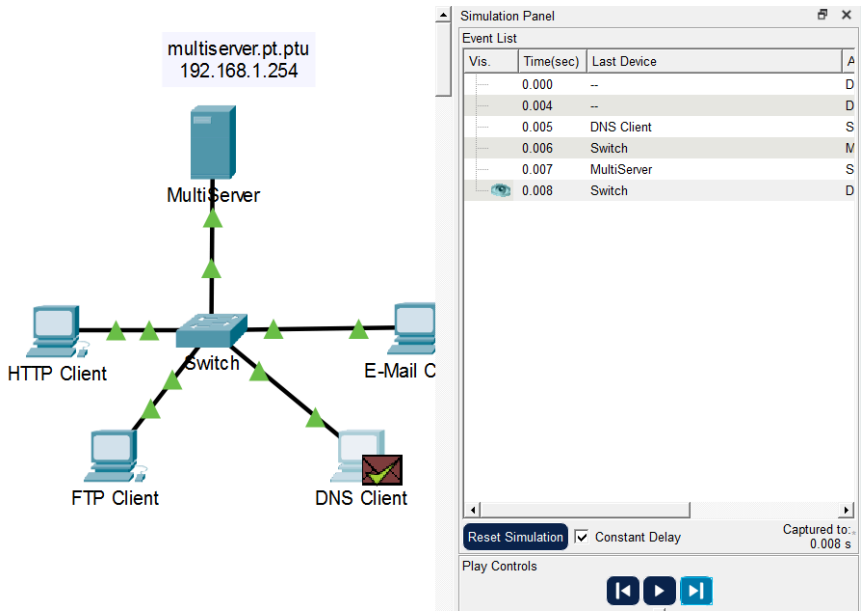
DEST PORT = 192.168.1.254

Question:

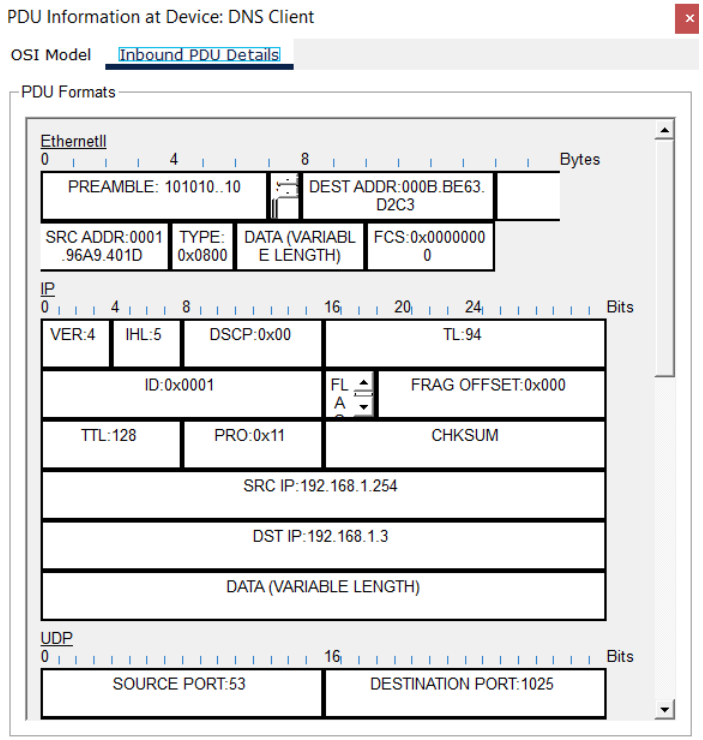
Why are there no sequence and acknowledgement numbers?

<u>UDP ไม่ใช้หมายเลขลำดับหรือหมายเลขยืนยัน</u>

f. Close the **PDU** and click **Capture/Forward** until a PDU with a check mark returns to the **DNS Client**.



g. Click the PDU envelope and select **Inbound PDU Details**.



Question:

How are the port and sequence numbers different than before?

SRC IP กับ DST IP สลับค่ากัน

What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

ส่วนสุดท้ายของ PDU เรียกว่า **DNS Answer (หรือ DNS Response)** ซึ่งเป็นส่วนที่แสดงผลลัพธ์ของการแปลงชื่อโดเมนเป็นหมายเลข IP หมายเลข IP ของชื่อ **multiserver.pt.ptu** คือ 192.168.1.254

h. Click Reset Simulation.

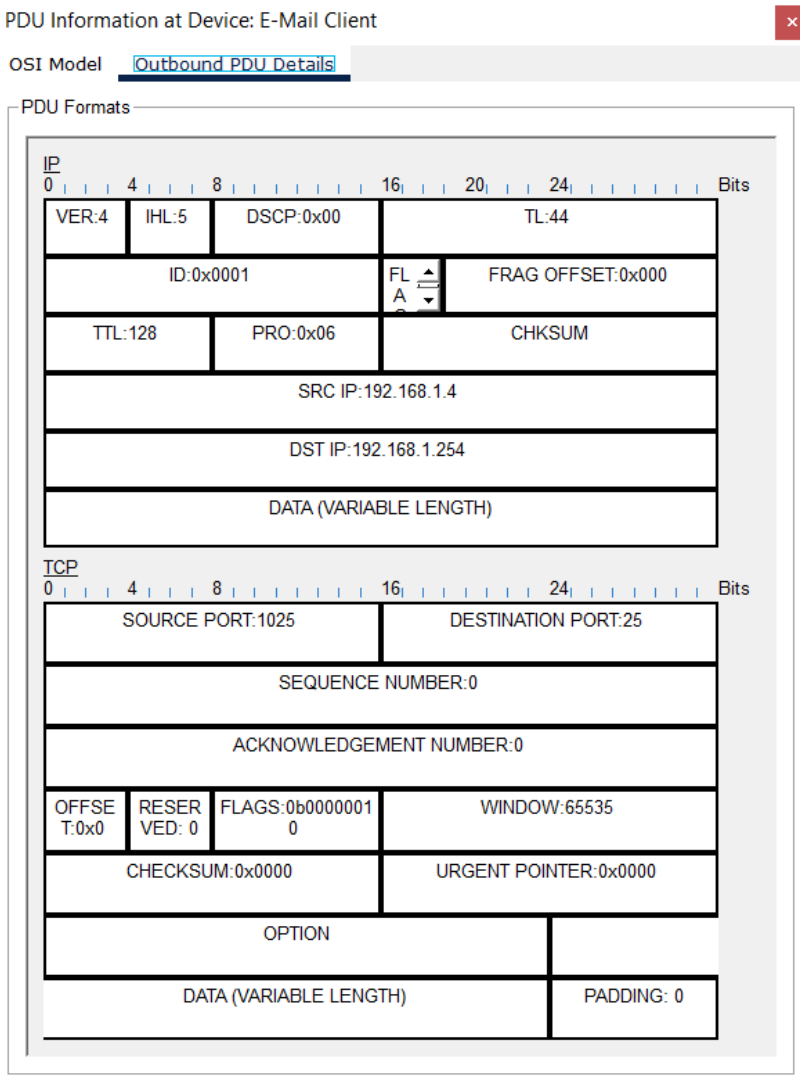
Step 4: Examine email traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to send an email to **user@multiserver.pt.ptu**.

b. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.

c. Click the first PDU envelope to open it.

d. Click the **Outbound PDU Details** tab and scroll down to the last section.



Questions:

What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

TCP เชื่อถือได้

e. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is the flag field value?

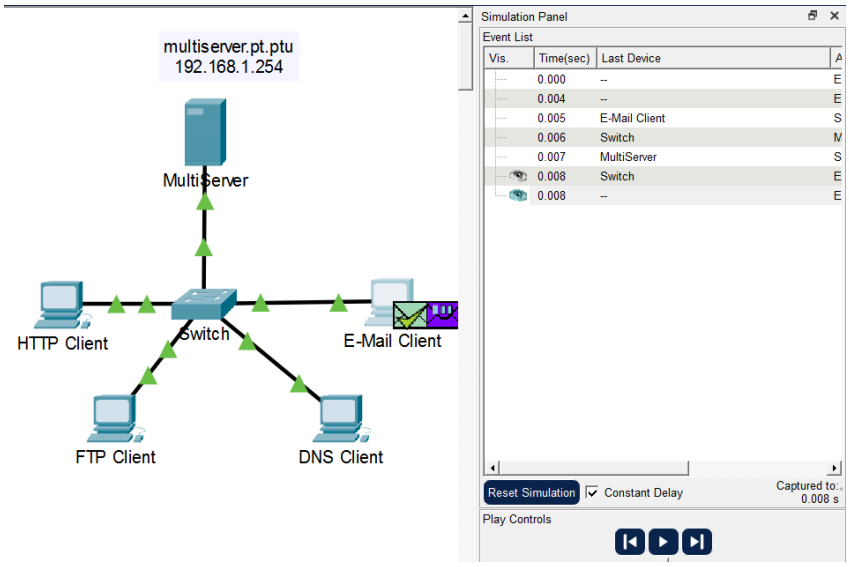
SRC PORT = 1025

DEST PORT = 25

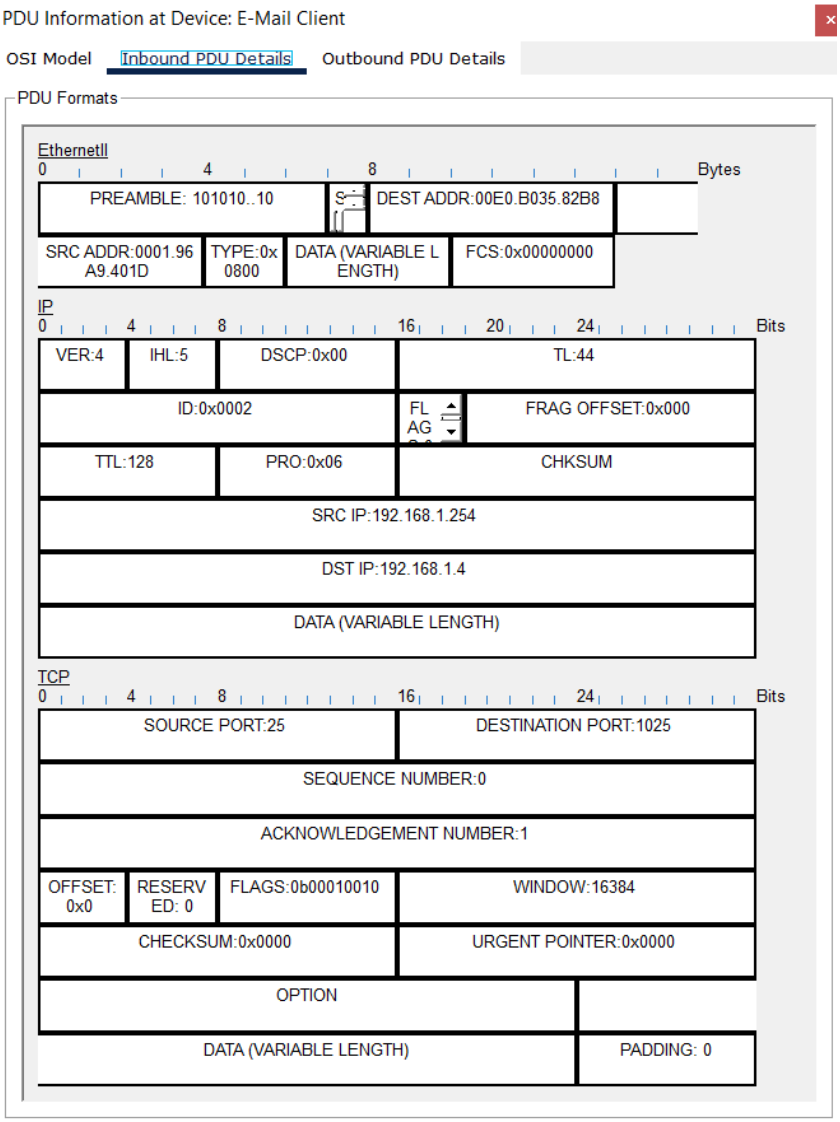
SEQUENCE NUM = 0

ACK NUM = 0

f. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.



g. Click the TCP PDU envelope and select **Inbound PDU Details**.

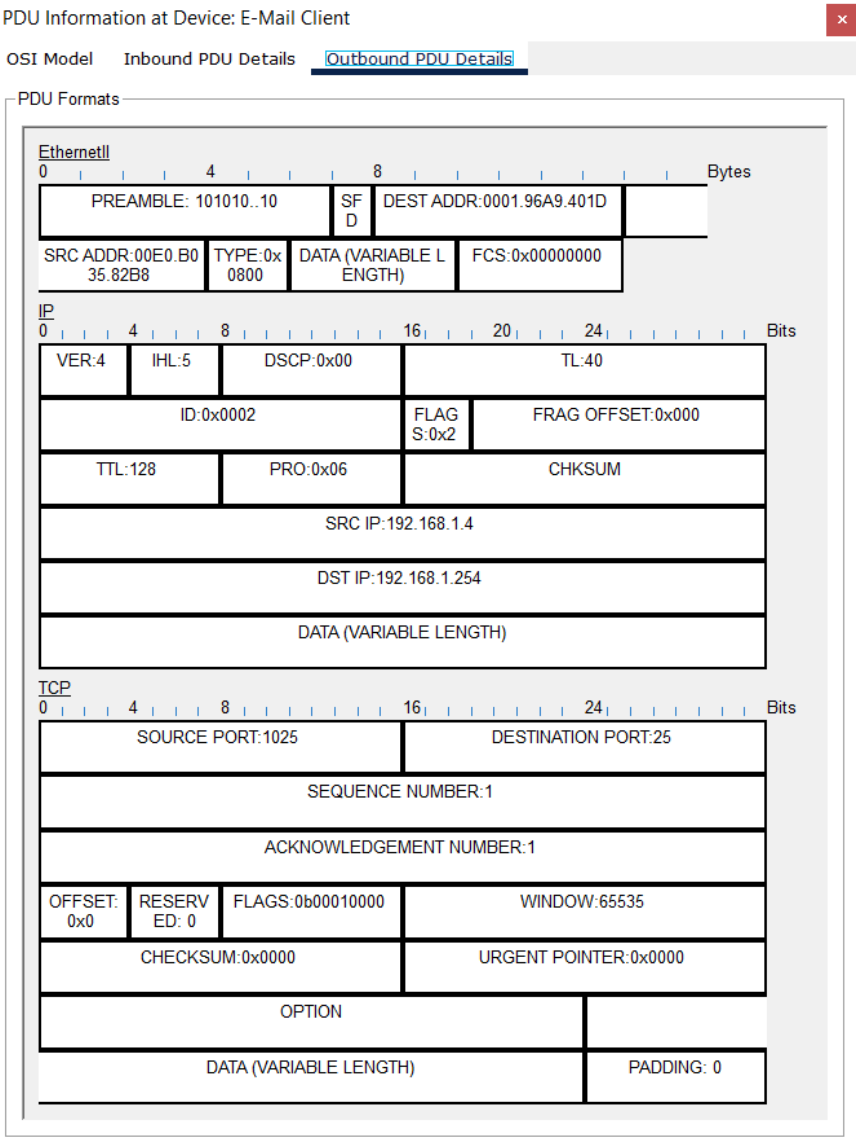


Question:

How are the port and sequence numbers different than before?

<u>ค่าของพอร์ตถูกสลับตำแหน่ง และหมายเลขยืนยันการรับส่งข้อมูล(ACK NUM) เปลี่ยนแปลงไป แต่หมายเลขลำดับ(SEQUENCE NUM)เหมือนเดิม</u>

h. Click the **Outbound PDU Details** tab.

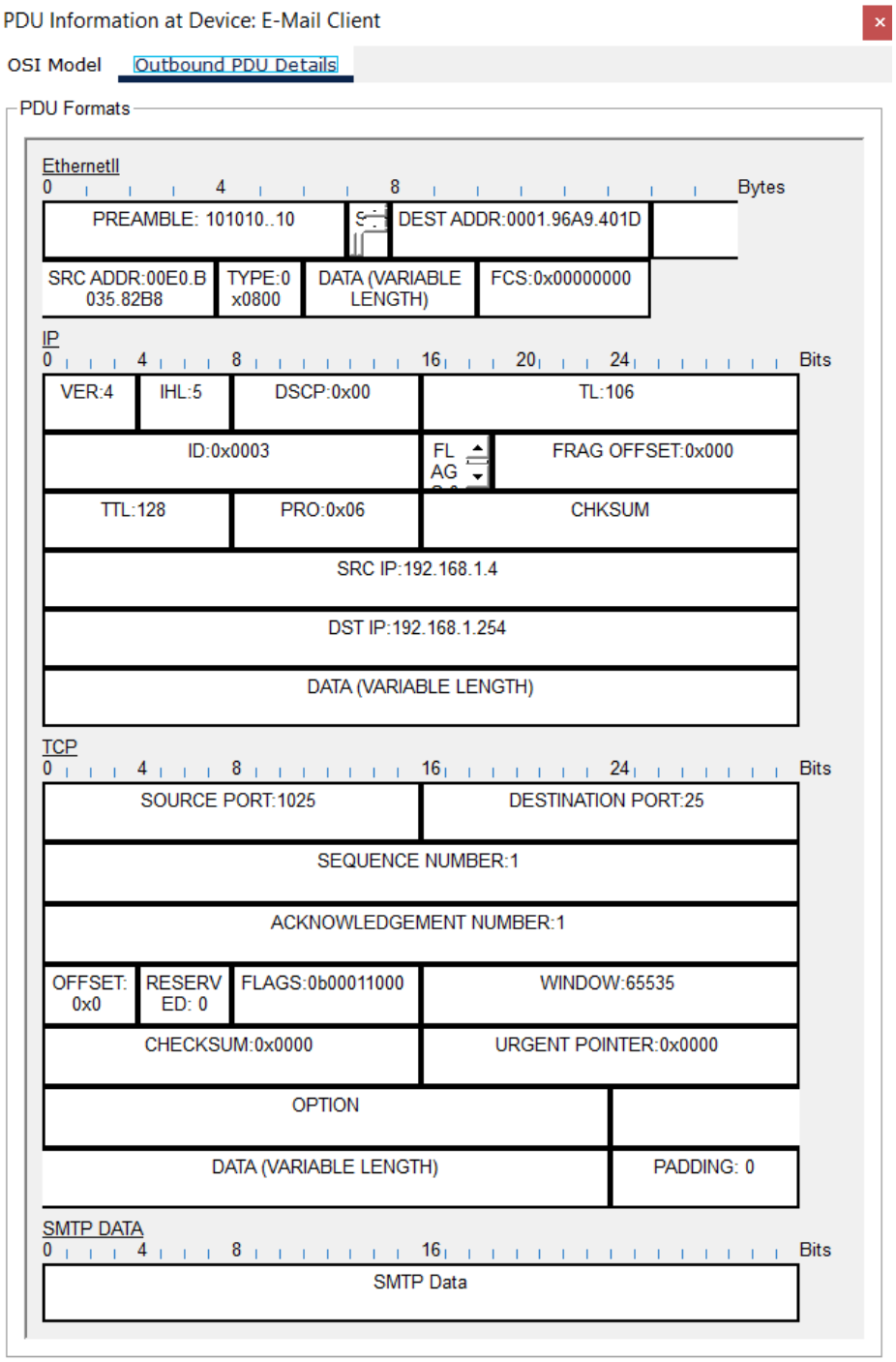


Question:

How are the port and sequence numbers different from the previous two results?

<u>ค่าของพอร์ตถูกสลับตำแหน่ง และหมายเลขลำดับ(SEQUENCE NUM) เปลี่ยนแปลงไป</u>

i. There is a second **PDU** of a different color that **E-Mail Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.



Questions:

How are the port and sequence numbers different from the previous two PDUs?

<u>เลขพอร์ต หมายเลขยืนยันการรับส่งข้อมูล(ACK NUM) และหมายเลขลำดับ(SEQUENCE NUM) เหมือน Outbound PDU Details ก่อนหน้า</u>

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

โปรโตคอลที่ใช้พอร์ต TCP 25 คือ SMTP

และโปรโตคอลที่ใช้พอร์ต TCP 110 คือ POP3